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I confirm that I understand my coursework needs to be submitted online via My Second Teacher under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

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1. Introduction

1.1 Introduction to the topic

- The concept of an online booking and tracking website concept came from the growing demand for efficient as possible, minimizing unnecessary obstacles and user-friendly method for booking tickets and tracking the travelling vehicle. In the growing world of digitalization, passengers also look for convenience and efficient way of booking ticket. Traditional methods such as buying tickets going at bus station or a ticket counter is very time consuming.

  Online platform could be a great solution, by offering features such as booking ticket from home and tracking vehicle, when it will reach at the stop. plan and manage your trips using these online tools. These websites give users the power to secure tickets by combining real-time tracking tools, but in context of Nepal there is no proper implementation of such websites. This project is being developed as a web application to solve these kinds of problems

1.2 Current Scenario

- Current Scenario in online ticket booking is that it includes seat displacement, lack of timely seat updates, not knowing about departure times and arrival times, inconvenience due to late departures, insufficient details about booked vehicles, and the absence of online reviews for passenger feedback. Furthermore, there's a lack of special discounts for students and senior passengers. Addressing these issues is crucial for improving the overall user experience in online ticket booking.

  There is no any tracking we application available current in Nepal which leads to very much hectic situation.

1.3 Project Scope

- The project scope of this project is developing a user-friendly online application to track bus and purchase tickets. It has benefits which includes safe payment processing, and real-time tracking. Passengers, service providers, support are all managed in the system. In order with a scheduled time for development, the goals includes scalability etc.

1.4 End User

i. Passengers: Person who want to track the location of their booked vehicles and looking to book tickets for transportation services online.

ii. Transportation Service Providers: Companies of bus services who are giving transportation services.

iii. Booking Agents: People who manages ticket bookings.

iv. Administrators: System administrators who manage and maintain the web application.

v. Customer Support: Peoples who are providing assistance and support to users in case of issues which are related to tracking or ticket bookings or any other issues.

1.5 Aims and Objectives

- The aims of this project is to create a online ticket booking and vehicle tracking site. This project aims to create real time GPS tracking system with an convenient ticket booking method where there will be the collaboration with both vehicle owner/vehicle staff and also with passengers. A user-friendly interface with the rating and review system is aimed to be build along with no misinformation and customer service.

- Objectives are :-

  i. Implement a real-time vehicle tracking system that provides accurate location updates between a period of time.
  ii. Enable passengers to select specific seats or preferences during the booking process (e.g., window seat).
  iii. Implement a feedback mechanism for vehicle owners/staff to respond to passenger inquiries within 24 hours.
  iv. Analyze reviews to identify common pain points and areas for improvement.
  v. Regularly update and maintain accurate information about schedules, pricing, and vehicle availability.
  vi. Create a portal for customer service

2. Project Background

2.1 Tools and Technology Used

i. HTML

- HTML stands for Hypertext Markup Language, which is a text-based language used to describe the structure of material inside HTML files and can make a web browser. A web browser can display text, images, and other multimedia on a webpage. (Lutkevich, 2020)

ii. CSS

- CSS is not a language used for programming. Additionally, it is not a markup language. A language used for style sheets is CSS. The method for styling HTML elements selectively is CSS. (CSS basics, n.d.)

iii. JS[Library-React]

- Developed with precision by Facebook, React.js is an open-source JavaScript library designed to make the complex process of creating user interfaces easier to understand. Using React, imagine creating a user interface that is of various components, each of which generates a short, reusable HTML code. (herbert, 2023)

iv. NodeJS

- An open-source, cross-platform runtime environment for JavaScript programming is called Node.js (Node). Developers didn't have to learn a second language to write client-side and server-side JavaScript code because Node is widely used for server-side programming. (Denman, n.d.)

v. MYSQL

- MySQL is like a helpful tool that organizes information in a way that makes it easy for computers to find and use. People use it a lot for different things, from small projects to big websites and business solutions. It's like a smart organizer for data. (G, 2024)

vi. Socket.io

- Socket.io is a JavaScript tool that helps different parts of a website or application talk to each other in real-time. It has two sides: one that works in web browser and another for a server using Node.js. They both use the same set of rules to communicate, making it easy for them to understand each other. This helps create interactive and real-time features on websites or apps.

2.2 Project Development Technology

Rational Unified Process (RUP)

- Four separate phases make up the project life cycle of the Rational Unified Process Methodology (RUP), an agile software development methodology. These stages cover a variety of tasks, such as application, testing, implementation, analysis, and design, as well as modeling. The distinctive feature of RUP is its iterative character, which suggests that essential tasks are reviewed and carried out again during the course of the project, enhancing its flexibility and agility. (Janse, 2019)

RUP method is great for building online tracking and ticket booking websites because it lets us improve things step by step, ensuring a well-organized and flexible development process. It helps us communicate better, manage risks, and focus on creating a strong

foundation for our website from the beginning. RUP's adaptability and attention to quality also make it suitable for handling changes smoothly and delivering a reliable and efficient system.

2.3 Review of similar Project

2.3.1 Similar Project

- Founded in 2009, eSewa has become a well-known name in Nepal. They are country's first digital payment service provider, authorized by the Central Bank of Nepal, and certified under ISO 27001:2013. As a part of F1Soft International, a prominent FinTech company in Nepal, eSewa is committed to making a cashless economy a reality. (esewa, n.d.). But the technology of tracking the vehicle is not implemented.

Figure 1 Logo of esewa

2.3.2 Comparison Between System

S.N	Features	SAHAJ YATRI	ESEWA
1.	Login	Available	Available
2.	Register	Available	Available
3.	Track Vehicle Location	Available	Not Available
4.	Rating	Available	Not Available
5.	Feedback	Available	Not Available
6.	Offline Payment	Available	Not Available
7.	Online Payment	Available	Available
8.	Transaction History	Available	Available
9.	Show Available Vehicle	Available	Available
10.	Check Vehicle Function	Available	Not Available
11.	Book Vehicle	Available	Available

Table 1 Comparison between system

3. Project Development

3.1 Requirement Gathering

- The requirement of this project is gathered by conducting survey at bus park and online survey with public by the help of different facebook pages and also consulting with the vehicle operator at the bus park. While conducting online survey with public, different ideas was collided to form a nice platform.

3.2 Use Case

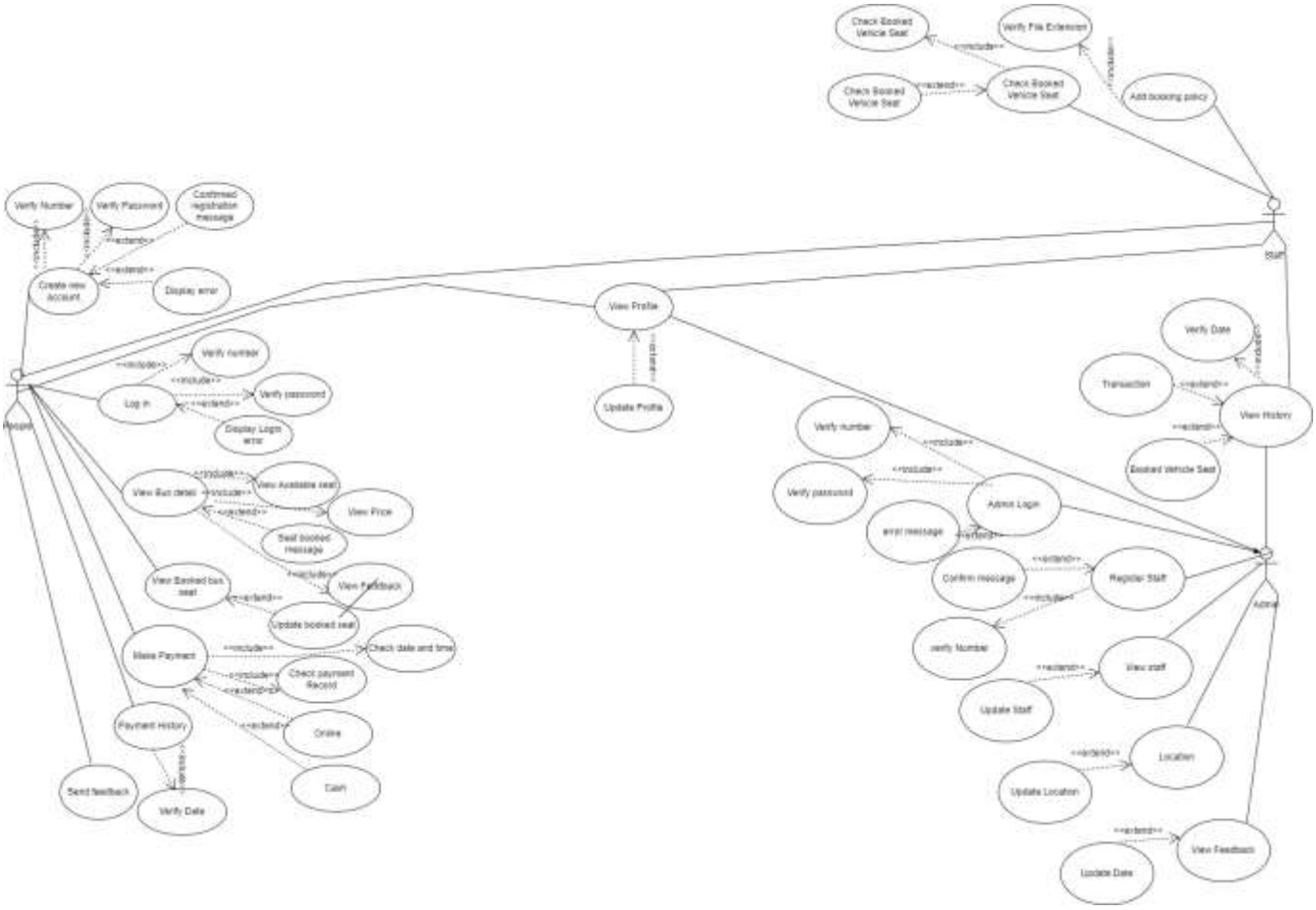

Figure 2 Use case diagram

3.3 SRS Document

- The full description of SRS document is kept in Appendix section.

3.4 Wire frames

- The images of wireframes are kept in Appendix section with its full description

3.5 ERD

- The image of ERD is in appendix.

4. Analysis of progress

- The analysis of progress includes the progress table, progress timeline and action plan which is presented below :-

4.1 Progress Table

SN	Tasks	Status	Progress (%)
1.	Topic Selection	Completed	100%
2.	Feasibility Study	Completed	100%
3.	Scheduling Resources	Completed	100%
4.	Cost and time estimation	Completed	100%
5.	Research on similar projects	Completed	100%
6.	Finalize Proposal	Completed	100%
7.	Conduct Public Survey	Completed	100%
8.	Conduct Survey at bus station with bus operators	Completed	100%

9.	Design Logo for the project	Completed	100%
10.	Develop Overall System Use Case Diagram	Completed	100%
12.	Develop SRS Document	Completed	100%
13.	Develop High Level Use Case Diagram	Completed	100%
14.	Develop Wireframes	Almost Completed	80%
15.	Finalized Initial ERD	Completed	100%
16.	Develop System Architecture Diagram	Completed	100%
17.	Develop Admin Panel in Web Application	Partially Completed	10%
18.	Finalize Interim Report	Completed	100%
19.	Develop System Operation Manual	Not Completed	0%
20.	Prepared Test Cases and Testing	Not Completed	0%
21.	Finalize SYP Report	Not Completed	0%
22.	Review and Refinement	Not Completed	0%

Table 2 Progress Table

4.2 Progress Timeline

- The works of the project has been carried out on almost the time according to the Gantt Chart which was submitted in the proposal.

5. Future Work

- The upcoming work involves completing the remaining tasks of the project in the future, with a timeline included in a Gantt Chart.

i. Create the web application's admin panel
Make an admin panel to manage employees, update vehicle, transaction tracking, and user reviews.

ii. Make an Operational Manual for the System:
Create a guide that explains how to make payments both online and offline, and how to reserve seats online, track the vehicle.

iii. Create test cases, then carry out the testing:
Create and run test cases to find and fix system flaws and guarantee a bug-free system.

iv. Complete the SYP report:
Finish the SYP report by adding the completed test cases, ERD, system architecture diagrams, sequence diagrams, and use case diagrams.

v. Examine and improve the project report:
The SYP report will be examine by supervisor and the feedback will be implemented in order to improve it and create properly formatted documentation.

6. References

(n.d.). Retrieved from esewa: https://blog.esewa.com.np/about-esewa/

(n.d.). Retrieved from https://www.sciencedirect.com/topics/computer-science/rational-unified-process#:~:text=To%20address%20risk%2C%20RUP%20organizes,elaboration%2C%20construction%2C%20and%20transition.

CSS basics. (n.d.). Retrieved from MDN web docs: https://developer.mozilla.org/en-US/docs/Learn/Getting_started_with_the_web/CSS_basics

Denman, R. S. (n.d.). *Node.js(Node)*. Retrieved from TechTarget: https://www.techtarget.com/whatis/definition/Nodejs#:~:text=js%20(Node)%3F-,Node.,to%20learn%20an%20additional%20language.

G, D. (2024, 1 19). *what is mysql*. Retrieved from Hostinger: https://www.hostinger.com/tutorials/what-is-mysql

herbert, D. (2023, 11 13). *Introduction to javascript*. Retrieved from hubspot: https://blog.hubspot.com/website/react-js

Janse, B. (2019, 08 16). *RUP*. Retrieved from toolshero: https://www.toolshero.com/information-technology/rational-unified-process-rup/

Lutkevich, B. (2020, 02). *HTML*. Retrieved from The serverside: https://www.theserverside.com/definition/HTML-Hypertext-Markup-Language

RUP overview. (n.d.). Retrieved from study.com: https://study.com/learn/lesson/rational-unified-process-methodology-examples.html#:~:text=Rational%20Unified%20Process%20(RUP)%20is%20considered%20an%20agile%20software%20development,requirements%20and%20objectives%20are%20met.

7. Appendix

7.1. SRS Document

7.1.1. Project Title :- Sahaj Yatri

7.1.2. Category :- Web application

7.1.3. Introduction

The Software Requirements Specification (SRS) document is created to offer comprehensive insights into the functions and aspects that the system must or must not have. It outlines the necessary features for both the Android and web applications. Additionally, it provides detailed information regarding the hardware, software, and any other technical requirements that the system depends on.

Project Scope

The project's scope involves creating a user-friendly online application dedicated to bus tracking and ticket purchasing. The key advantages are secure payment processing and real-time tracking functionalities. The system manages passengers, service providers. The development

timeline address to a well-planned schedule, aiming to achieve goals such as scalability. The main objective is to provide a efficient solution that enhances the overall experience for users involved in bus transportation, ensuring convenience, safety, and reliability throughout the journey.

Existing System

- The existing system is Founded in 2009 which is esewa. eSewa has become a well-known name in Nepal. They are country's first digital payment service provider, authorized by the Central Bank of Nepal, and certified under ISO 27001:2013. As a part of F1Soft International, a prominent FinTech company in Nepal, eSewa is committed to making a cashless economy a reality. (esewa, n.d.). But the technology of tracking the vehicle is not implemented.

  But esewa do not have facility to track the vehicle.

7.2. Wireframes

i. View Page

- This is the page which is opened at first when a person enter at the website from where customer can login their account or can create their account if they are new to the website.

Figure 3 first wireframe

ii. Login Page

- This is the login page for both customers and staff members. Here, users can input the email and password they used when signing up for the Sahaj Yatri. Once logged in, they can access all the features offered by the app.

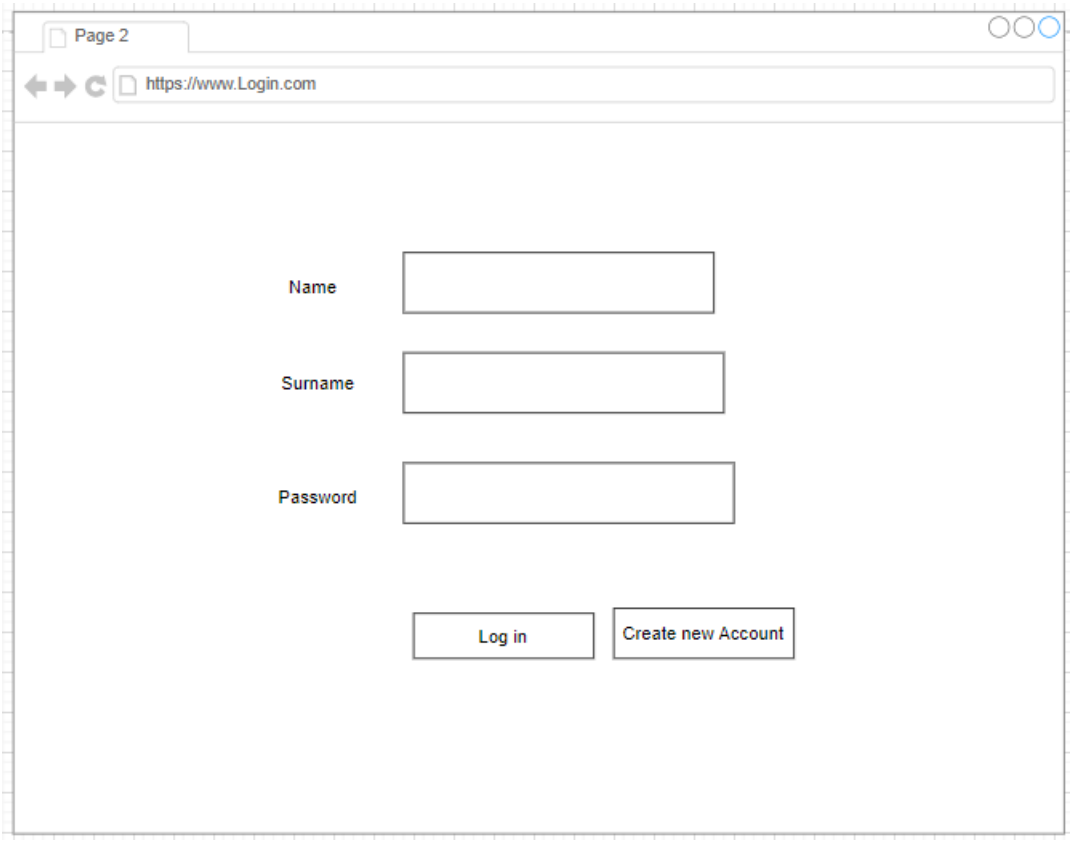

Figure 4 Login Page

iii. Create Account Page

- This is the create account page for every public. If a customer is looking to use the Sahaj Yatri, they need to register first by completing the provided form.

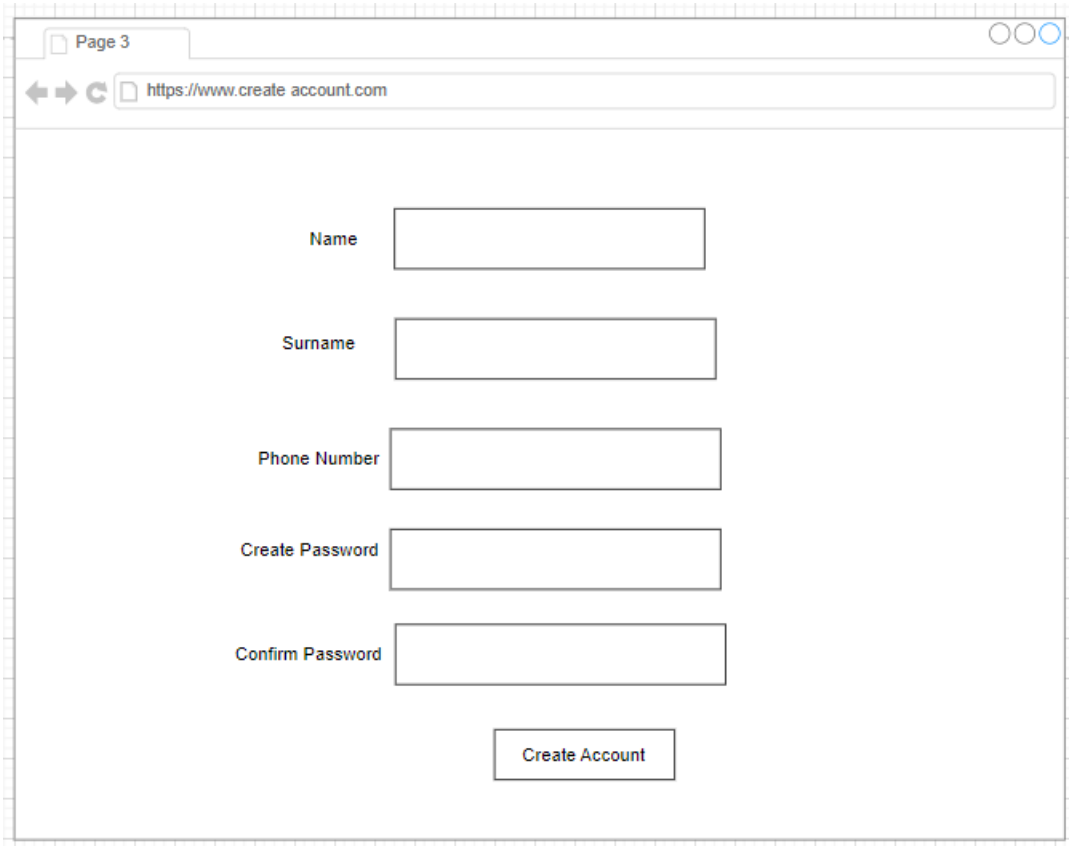

Figure 5 Create Account Page

iv. Search Vehicle Page

- When the user will login their account then they will directly reach to this page from where they can select their city, desired destination and the date when they are desiring to departure.

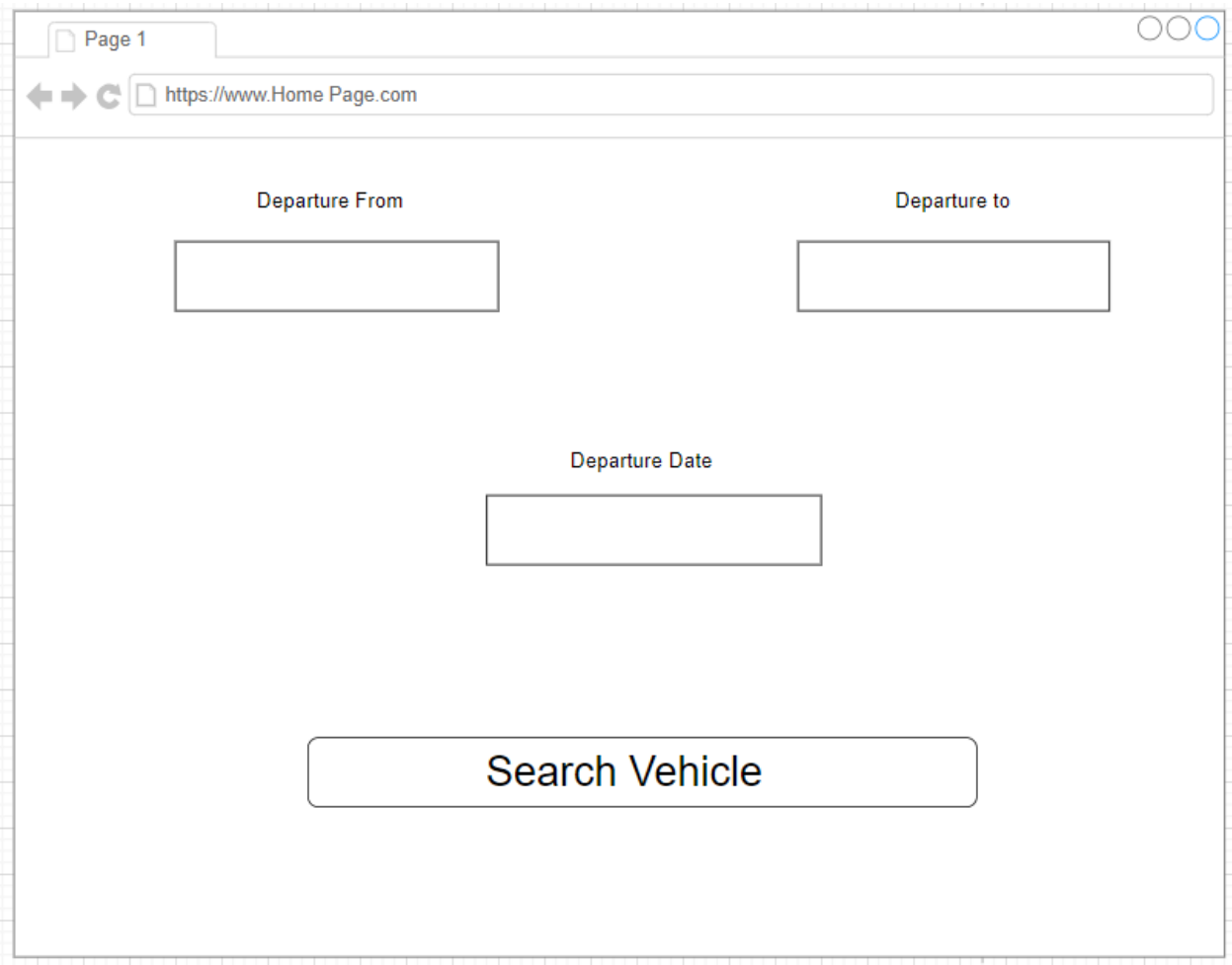

Figure 6 Search Vehicle Page

v. Bus Detail Page

- When the user will click in search vehicle then they will directly reach to this page from where they can see the available vehicle details and departure time of the vehicle and select according to it.

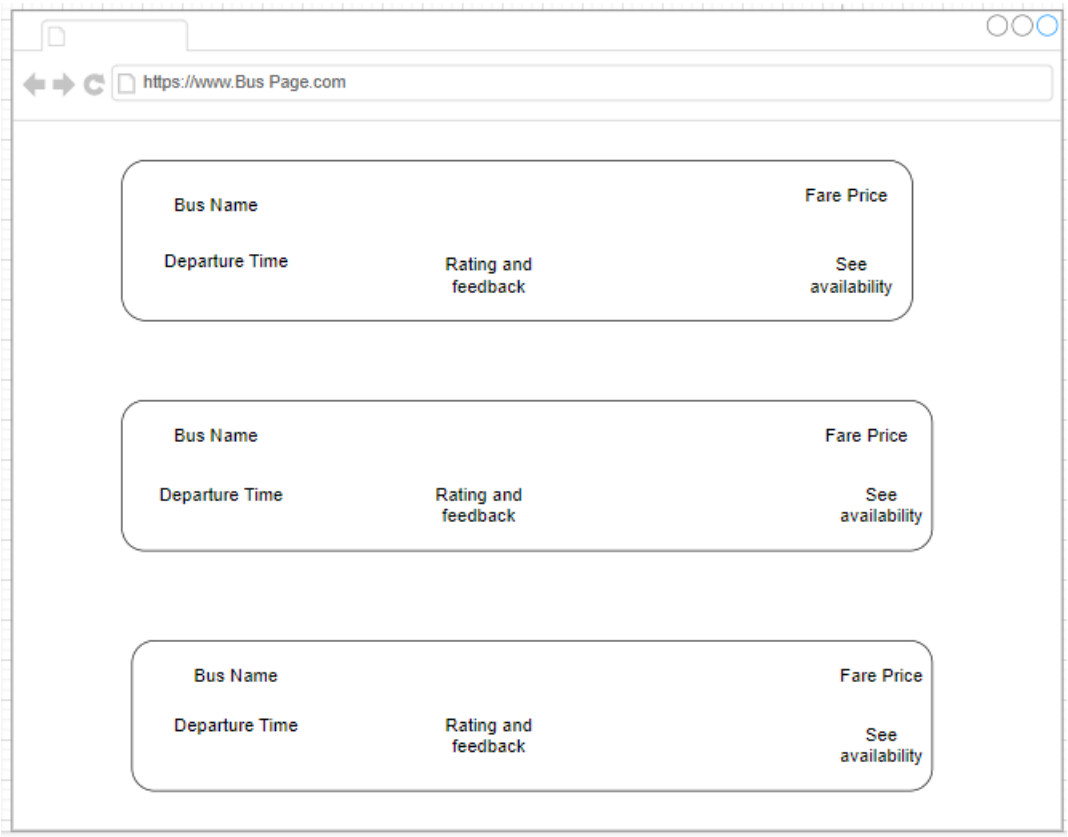

Figure 7 Bus Detail Page

vi. Book Vehicle Page

- When the user will click in see availability then they will directly reach to this page from where they can see the available vehicle seats and can select the seat according to their wish and can finally book their vehicle.

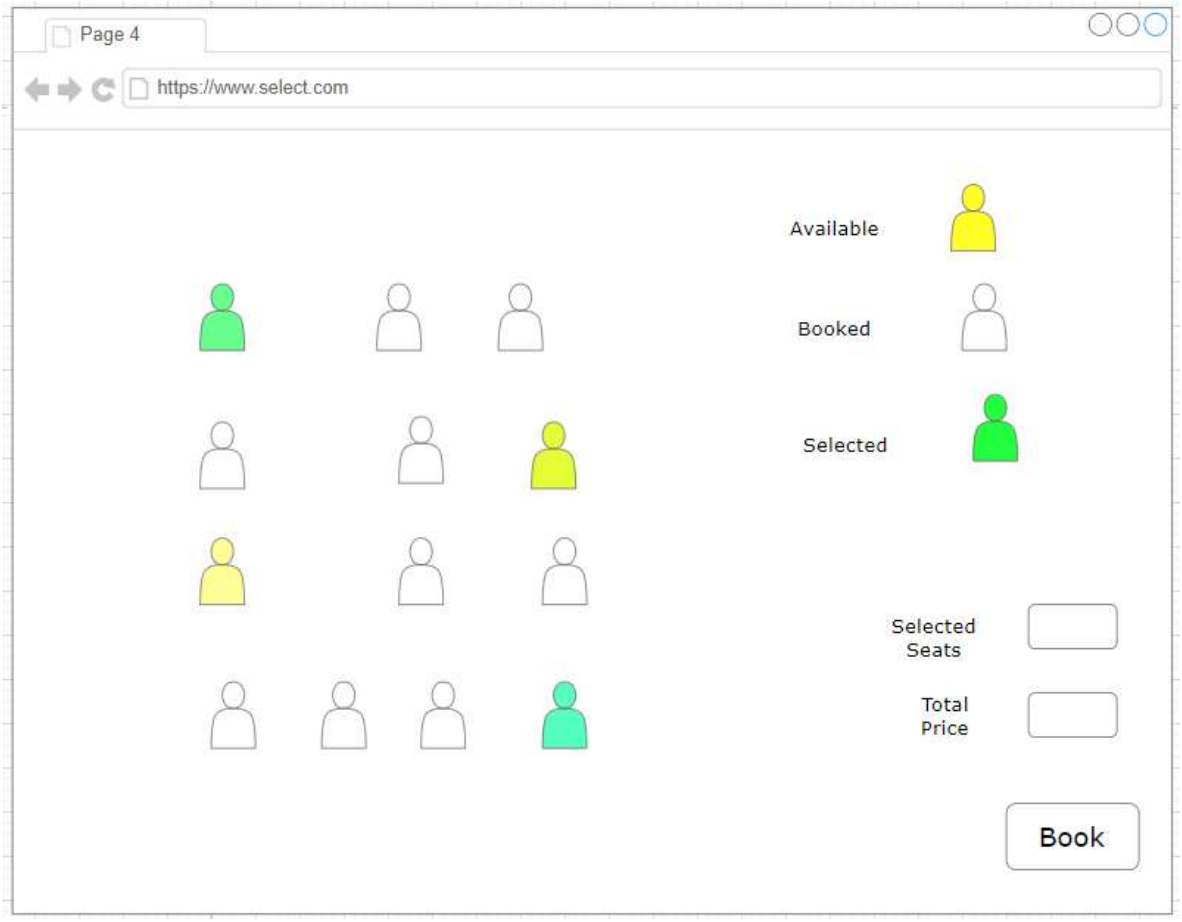

Figure 8 Book Vehicle Page

vii. Feedback Page

- From feedback page user can send feedback directly to the admin about the vehicle and admin will then consult with the owner.

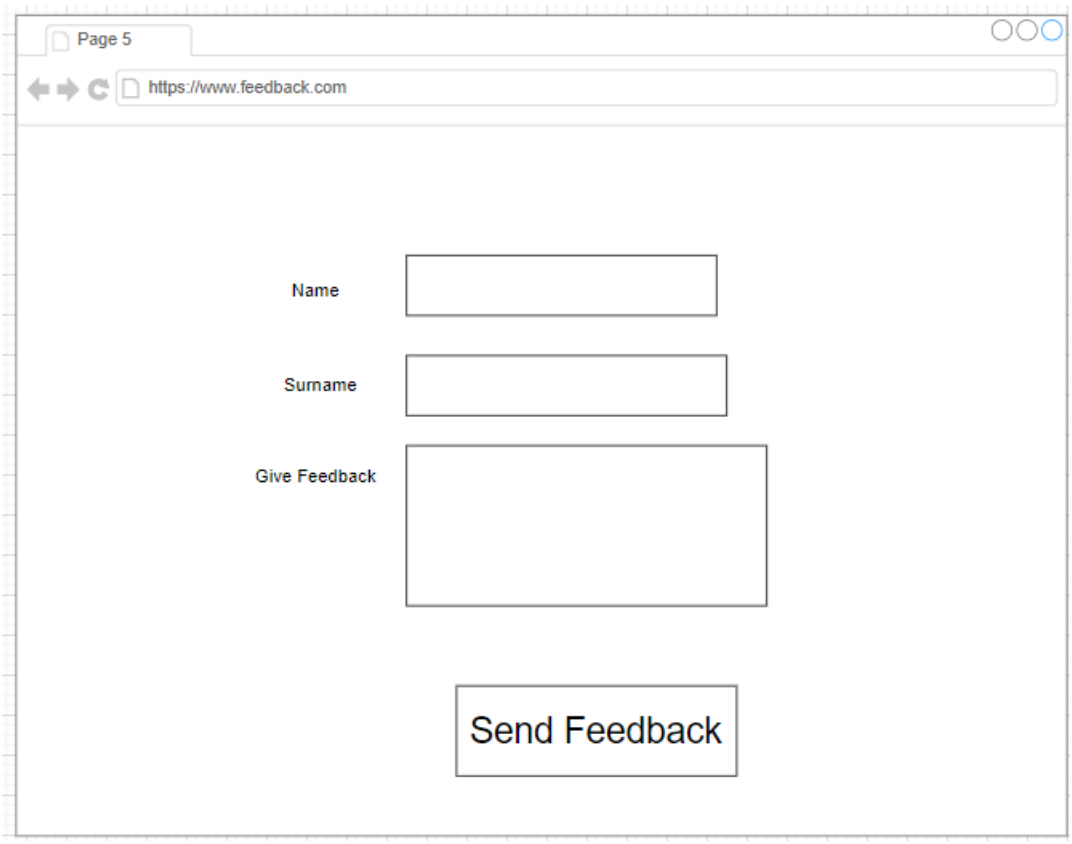

Figure 9 Send feedback Page

7.3. Entity Relationship Diagram

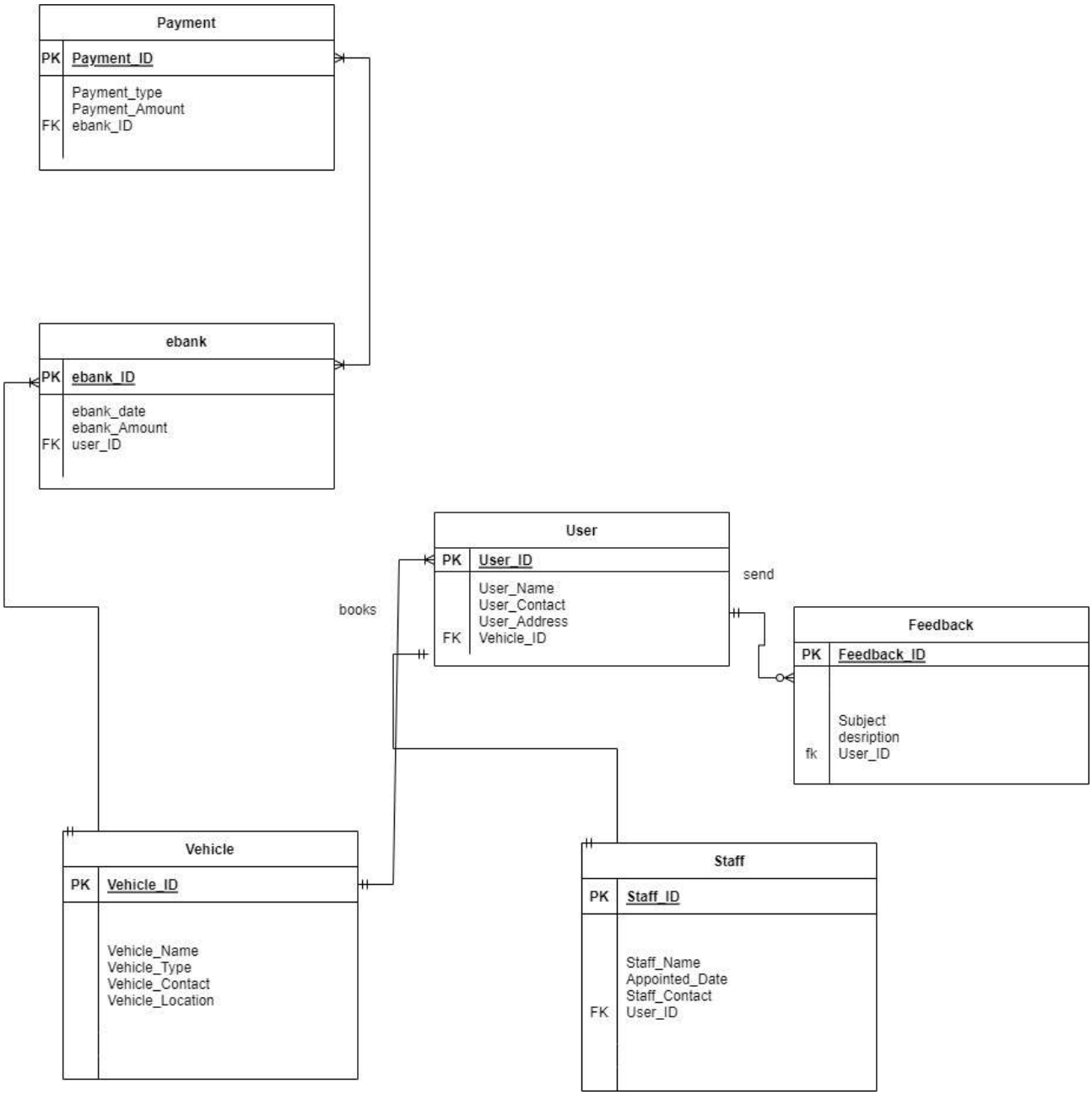

Figure 10 Entity Relationship Diagram

7.4. Project as a Solution

- Passengers do not have to worry about seat misplacement, as the web application prevents such errors. Seat information gets updated time to time after bookings or cancellations after being ensured about no duplicate bookings.

  Passengers can have accurate departure times from their stations. With a real-time tracking system in place, if there is any delay, passengers can track the vehicle and get arrival times.

  The website provides accurate details about the vehicle, and all the promised facilities are genuinely offered to passengers. Dissatisfied passengers have the option to leave reviews, which will help future travelers assess the ratings and reviews of the vehicles they plan to use.

  Special discounts are available for student passengers and those above 70 years old, a feature not commonly found on other online platforms or at ticket counters. This initiative aims to make travel more affordable for these specific passenger groups.

7.5. Project Gantt Chart

7.6. Selected Methodology

Rational Unified Process (RUP)

- Four separate phases make up the project life cycle of the Rational Unified Process Methodology (RUP), an agile software development methodology. These stages cover a variety of tasks, such as application, testing, implementation, analysis, and design, as well as modeling. The distinctive feature of RUP is its iterative character, which suggests that essential tasks are reviewed and carried out again during the course of the project, enhancing its flexibility and agility. (Janse, 2019)

i. Inception

- Team talks about the project's basic concept, structure, and resources are conducted during the genesis phase of the RUP approach. Research on market trends, financial projections, project scheduling, risk evaluations, vision or scope statements, and, in certain situations, prototypes are all part of this phase. (RUP overview, n.d.)
- Inception phases for my project consists of :-
  a. Finalizing initial idea of the project
  b. Determination of which language is to be used
  c. What feature will be there
  d. Title defence
  e. Discussion with supervisor about development and documentation

ii. Elaboration

- The goal of the elaboration phase in the RUP approach is to further minimize associated risks by conducting a more thorough examination of the development plan. When being utilized, prototypes are developed and the necessary system architecture is evaluated to make it possible. (RUP overview, n.d.)
- Elaboration phases for my project consists of :-
  a. Time evaluation
  b. Risk assessment
  c. Construct WBS and Gantt chart
  d. Documentation of proposal is created

iii. Construction

- Project development and completion take place during the construction phase of the RUP process. This stage encompasses the creation of the system's source code as well as other features. (RUP overview, n.d.)
- Construction phases for my project consists of :-
  a. Coding
  b. Testing of the project
  c. Documentation of the project

iv. Transition

- In transition phase, an application that has been completed is handed over to the operations team for management, upkeep, and support. (htt)

- Transition phases for my project consists of :-

a. Analysis of user review
b. Maintain Product
c. Transformation of current database